Book 1 — Lecture 2 — Exercise 3

Chain Rule for Three Composed Functions

Prompt. Use the chain rule to find the derivatives of each of the following functions.[1]

$$g(t) = \sin(t^2) - \cos(t^2),$$
$$\theta(\alpha) = e^{3\alpha} + 3\alpha\ \ln(3\alpha),$$
$$x(t) = \sin^2(t^2) - \cos(t^2).$$

$g(t) = \sin(t^2) - \cos(t^2)$

Set $u(t) = t^2$, so $\dfrac{du}{dt} = 2t$.[2] Then $g(t) = \sin u - \cos u$, and

$$\frac{dg}{dt} = \cos u \cdot \frac{du}{dt} - (-\sin u) \cdot \frac{du}{dt} = (\cos u + \sin u)\frac{du}{dt}.$$

Substitute $u = t^2$:

$$g'(t) = 2t\big(\cos(t^2) + \sin(t^2)\big).$$

$\theta(\alpha) = e^{3\alpha} + 3\alpha\ \ln(3\alpha)$

Require $\alpha > 0$ so that $\ln(3\alpha)$ is defined.[3]

For $e^{3\alpha}$, let $v = 3\alpha$; then $\dfrac{d}{d\alpha}e^{3\alpha} = e^v \cdot 3 = 3e^{3\alpha}$.

For $3\alpha\ \ln(3\alpha)$, use the product rule with $a(\alpha) = 3\alpha$ and $b(\alpha) = \ln(3\alpha)$.[4] Now $a' = 3$ and $b' = 1/\alpha$, so

$$\frac{d}{d\alpha}\big(3\alpha\ \ln(3\alpha)\big) = 3\ln(3\alpha) + 3\alpha \cdot \frac{1}{\alpha} = 3\ln(3\alpha) + 3.$$

Therefore

$$\theta'(\alpha) = 3e^{3\alpha} + 3\ln(3\alpha) + 3 = 3\big(e^{3\alpha} + \ln(3\alpha) + 1\big).$$

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[1] The *chain rule*: if $y = h(u)$ and $u = k(t)$ are differentiable, then $\dfrac{dy}{dt} = h'(u)\,k'(t)$. The same idea applies when the outer function is exponential, logarithmic, or built from sine and cosine.

[2] Here the "inner" function is t^2; every appearance of t^2 in sin and cos is differentiated through this factor $2t$.

[3] The argument of ln must be positive: here $3\alpha > 0$, equivalently $\alpha > 0$.

[4] $\dfrac{d}{d\alpha}\ln(3\alpha) = \dfrac{1}{3\alpha} \cdot 3 = \dfrac{1}{\alpha}$ by the chain rule on $\ln(3\alpha)$.

$$x(t) = \sin^2(t^2) - \cos(t^2)$$

Write $\sin^2(t^2) = \big(\sin(t^2)\big)^2$. Again set $u = t^2$ with $\dfrac{du}{dt} = 2t$.[5] First,

$$\frac{d}{dt}\sin^2(t^2) = 2\sin(t^2)\cos(t^2) \cdot \frac{d}{dt}(t^2) = 4t\ \sin(t^2)\cos(t^2).$$

Also $\dfrac{d}{dt}\big(-\cos(t^2)\big) = \sin(t^2) \cdot 2t = 2t\sin(t^2)$. Hence

$$x'(t) = 4t\ \sin(t^2)\cos(t^2) + 2t\sin(t^2) = 2t\ \sin(t^2)\big(2\cos(t^2) + 1\big).$$

Result. Summarizing (with $\alpha > 0$ for θ),

$$\begin{aligned} g'(t) &= 2t\big(\cos(t^2) + \sin(t^2)\big), \\ \theta'(\alpha) &= 3e^{3\alpha} + 3\ln(3\alpha) + 3, \\ x'(t) &= 2t\ \sin(t^2)\big(2\cos(t^2) + 1\big). \end{aligned}$$

Example. At $t = 0$: $g'(0) = 0$ and $x'(0) = 0$ because every term carries a factor t. At $\alpha = 1$: $\theta'(1) = 3e^3 + 3\ln 3 + 3$.

[5] This combines two uses of the chain rule: squaring the sine, and the argument t^2 inside the sine.